```
> restart;#GyroMag
```

```
> mp := 1.672621923·10^-27; hbar := 6.62607015/(2·Pi)·10^-34; B2 := 4.2·10^14; ec := 1.602176·10^-19; 1/ec;
```

$$mp := 1.672621923 \times 10^{-27}$$

$$\hbar := 1.054571818 \times 10^{-34}$$

$$B2 := 4.200000000 \times 10^{14}$$

$$ec := 1.602176000 \times 10^{-19}$$

$$6.241511544 \times 10^{18} \quad (1)$$

```
> γ0 := 9.57865·10^7; I0 := 1/2; γ0 := 10.01926·10^7; I0 := 1/2; γ0 := 4.81519·10^7; I0 := 1/2; γ0 := -1.77748
  ·10^7; I0 := 3/2;#γ0 :=5.64661·10^7; I0 := 1/2;
```

$$\gamma 0 := 9.578650000 \times 10^{7}$$

$$I0 := \frac{1}{2}$$

$$\gamma 0 := 1.001926000 \times 10^{8}$$

$$I0 := \frac{1}{2}$$

$$\gamma 0 := 4.815190000 \times 10^{7}$$

$$I0 := \frac{1}{2}$$

$$\gamma 0 := -1.777480000 \times 10^{7}$$

$$I0 := \frac{3}{2} \quad (2)$$

```
> EeV := hbar·γ0·I0·(1 - I0)·B2/ec;#Energy eV Sn117=6.620023680×10^6; Sn119 = 6.924539308 × 10^6;
  Hg 199 = 3.327887733 × 10^6; Hg201 = 3.685371015 × 10^6; Pb207 = 3.902501074 × 10^6
```

$$EeV := 3.685371015 \times 10^{6} \quad (3)$$

```
> EJ := hbar·γ0·I0·(1 - I0)·B2;# EJ Joule Sn117 = 1.060644306 × 10^-12; Sn119 = 1.109433069 × 10^-12;
  Hg199 = 5.331861856 × 10^-13; Hg201 = 5.904612991 × 10^-13; Pb207 := 6.252493561 × 10^-13;
```

$$EJ := 5.904612991 \times 10^{-13} \quad (4)$$

```
> EJHg199201 := (5.331861856 × 10^-13·0.132 + 5.904612991 × 10^-13·0.168)·3.002213849 × 10^24;
  #Energy in Joule/Kg Hg natural abundance
```

$$EJHg199201 := 5.091096447 \times 10^{11} \quad (5)$$

```
> NA := 6.02214076 × 10^23;#Atoms per mol Hg= 0.20059Kg
```

$$NA := 6.022140760 \times 10^{23} \quad (6)$$

```
> HgmolKg := NA/0.20059
```

$$HgmolKg := 3.002213849 \times 10^{24} \quad (7)$$

```
> EJSnPb117207 := (1.065433069 × 10^-12·0.166·4.751 ·10^22 + 6.252493561 × 10^-13·0.221·2.333·10^22);
  # with the aproximation of Energy Sn(117+119)= 1.065433069×10^-12 J
```

$$EJSnPb117207 := 1.162645028 \times 10^{10} \quad (8)$$

```
>
```